

Sewade Ogun

Paris, France

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Summary

AI Researcher specializing in conversational AI systems. I design and build the end-to-end architecture for next-generation voice agents ranging from native speech-to-speech (S2S) models to optimized cascaded pipelines across Speech-to-Text (STT), Large Language Models (LLMs), and Text-to-Speech (TTS) models to achieve seamless, low-latency human-machine interaction.

Work Experience

AI Researcher

Paris, France

GETVOCAL AI

May. 2025 - Present

- Research, development, and integration of speech-to-text and text-to-speech models into the voice agent pipeline, particularly for low-resource (European) languages.
- Research on other aspects of speech-to-speech models, like end-of-turn detection, intent understanding, code-switching, latency reduction, etc.

Researcher

BIO-RAMP RESEARCH LAB | MASAKHANE

Community

- Research and develop STT/ASR and TTS models for low-resource languages, pioneering data-efficient training methodologies to bridge the performance gap in under-represented languages.
- Developed the first Text-to-Speech (TTS) model for African-accented English, mitigating regional data scarcity to synthesize highly natural, localized, and anonymised voices.
- Developed the first state-of-the-art multilingual Speech-to-Text (STT) model for Nigerian languages, with 30% average reduction in word-error-rate over monolingual baselines.

Ph.D Student

Nancy, France

INRIA

Oct. 2021 - Sept. 2024

- Designed generative multi-speaker text-to-speech (TTS) and voice conversion (VC) systems that substantially improved utterance quality, prosody, and speaker diversity.
- Developed robust weakly-supervised ASR models trained on hybrid real and synthetic datasets; mitigated dataset distributional shifts to improve word error rate (WER) by up to 40% on real-world test data.
- Developed and deployed an end-to-end plug-and-play vocal assistant framework at INRIA, seamlessly integrating ASR, LLMs, and TTS research outputs from the laboratory into a unified library.

Research and Teaching Assistant

Mbour, Senegal

AFRICAN INSTITUTE FOR MATHEMATICAL SCIENCES

Oct. 2020 - Oct. 2021

- Developed weakly-supervised and self-supervised models for learning speech representations across African languages
- Taught various machine learning courses including foundations of machine learning, natural language processing, among others

Machine Learning Research Engineer

Canada (Remote)

UBENWA

Mar. 2021 - August. 2021

- Developed and deployed accurate deep-learning models for baby cry activity detection and segmentation
- Developed models for detecting asphyxia from infant cry.

NLP Engineer

Canada (Remote)

PROTO

Feb. 2020 - Nov. 2020

- Built goal-oriented Conversational AI systems with Question Answering capabilities.

Predictive Analyst

Lagos, Nigeria

FLOUR MILLS OF NIGERIA PLC

Apr. 2017 - Oct. 2019

- Created visualization and analytics dashboards sourced from the factory production and logistics database to deliver actionable insights for informed decision-making.

Education

PhD. in Computer Science

Nancy, France

UNIVERSITY OF LORRAINE

Oct. 2021 - Oct. 2024

- Thesis: Generating diverse synthetic data for ASR training data augmentation
- Thesis supervisors: Emmanuel Vincent and Vincent Colotte

Masters in Mathematical Sciences (Machine Intelligence Specialization)

AFRICAN INSTITUTE FOR MATHEMATICAL SCIENCES (AIMS)

- Thesis: Towards a weakly-supervised learning paradigm for speech recognition
- Thesis supervisors: Moustapha Cisse and Yossi Adi
- Grade: Distinction

Masters in Electrical and Electronic Engineering

UNIVERSITY OF LAGOS

- Grade: Distinction

Bachelor of Technology in Electronic and Electrical Engineering

LADOKE AKINTOLA UNIVERSITY OF TECHNOLOGY

- Grade: First Class Honours

Ghana

Oct. 2019 - Oct. 2020

Lagos, Nigeria

2017 - 2019

Ogbomoso, Nigeria

2007 - 2012

Technical Skills

- Programming Languages and Libraries: Python, Javascript, NumPy, PyTorch, Tensorflow.
- Dev: SQL, Databases (PostgreSQL & MySQL), Caching, Asynchronous Programming
- Research: Research article writing, documentation, distributed computing, AI model development.
- Spoken Languages: English (Native), French (Fluent), Spanish (Intermediate).

Publications

Articles and Journals

- **S. Ogun**. Sometin Beta Pass Notin (SBPN): Improving Multilingual ASR for Nigerian Languages via Knowledge Distillation. To be submitted to Speech and Language Technologies Workshop (SLT), 2026. arXiv preprint arXiv:2605.17710
- **S. Ogun**, V. Colotte, and E. Vincent. Evaluating the importance of synthetic data augmentation in improving automatic speech recognition. Submitted to IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2024. Under review
- **S. Ogun**. Generating diverse synthetic data for ASR training data augmentation. PhD thesis, University of Lorraine, 2024. 2024.
- **S. Ogun**, A. Owodunni, T. Olatunji, E. Alese, B. Oladimeji, T. Afonja, K. Olaleye, N. Etori, T. Adewumi. 1000 African Voices: Advancing inclusive multi-speaker multi-accent speech synthesis. In Proc. Interspeech 2024. 2024.
- T. Afonja, T. Olatunji, **S. Ogun**, N. Etori, A. Owodunni, M. Yekini. Performant ASR models for medical entities in accented speech. In Proc. Interspeech 2024. 2024.
- G. Coiffier, **S. Ogun**, L. Valque, P. Trivedi. Think before loading (State of the art), 2024, 978-2-9591975-0-5
- **S. Ogun**, V. Colotte, and E. Vincent. Stochastic pitch prediction improves the diversity and naturalness of speech in Glow-TTS. In Proc. Interspeech, 2023, pp. 4878–4882.
- **S. Ogun**, V. Colotte, and E. Vincent. Can we use Common Voice to train a multi-speaker TTS system? In Proc. SLT. IEEE, 2023. pp. 900-905.
- A. Ndoeye, **S. Ogun**, Y. Adi, and M. Cisse. Towards a weakly-supervised learning paradigm for speech recognition. Masters thesis.

Blog articles

ARTICLES AVAILABLE AT [HTTPS://OGUNLAO.GITHUB.IO/ARCHIVE/INDEX.HTML](https://ogunlao.github.io/archive/index.html)

- Making Efficient Neural Networks.
- From GRU to Transformer.
- You Don't Really Know Softmax.
- Cross Validation and Reproducibility in Neural Network Training.

Certifications

- 2020 **Data Engineering with Google Cloud**, Google
- 2019 **Deep Learning Specialization**, deeplearning.ai
- 2019 **IBM Data Science Professional Certificate**, IBM
- 2014 **Oracle Database SQL Administration**, Oracle University

AI Research Groups

- Masakhane NLP Research Group: NLP research on low-resource languages
- Biomedical Research in Artificial Intelligence and Machine Perception (Bio-RAMP) Group: NLP and speech research for medical applications.